



Case Report

Role of Agnikarma along with Ayurveda medication in management of trigger thumb - A Case Report

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ABSTRACT

Trigger thumb is a condition caused by incongruity in diameter between the swollen flexor tendon and its sheath at the metacarpal head, often due to thickening of the A1 pulley. This results in a clicking or popping sensation during thumb extension. Treatment options include splinting, physical therapy, steroid injections, and surgical release of the A1 pulley. In Ayurveda, trigger thumb can be attributed to *Snayugata Vata* (vitiated *Vata* dosha in tendons/ligaments). *Agnikarma* is an Ayurvedic procedure, effective for treating *Snayugata Vata* and other joint disorders. A 56-year-old male patient, working in a printing press, presented with pain and swelling in his right thumb at National Institute of Ayurveda (NIA) hospital. On the basis of clinical examination, the patient was diagnosed with right trigger thumb. The patient underwent three sittings of *Bindu*-type *Agnikarma* therapy, using *Tamra Shalaka*, at site of maximum tenderness, at interval of seven days. Following treatment, the patient experienced complete resolution of symptoms with follow-up of five months. *Agnikarma* provided a successful, minimally invasive, and cost-effective alternative to surgical intervention, demonstrating its potential as an effective remedy for trigger thumb. This case study provides a lead to establish a new standardized treatment protocol and assess long term benefits of *Agnikarma* for trigger thumb.

1. Introduction

Trigger thumb (TT) is a musculoskeletal disorder of hand, represented by painful clicking sound produced when metacarpo-phalangeal joint flexed and extended. It occurs by an incongruity in diameter between the inflamed flexor tendon and A1 pulley at the metacarpal head level, leads to formation of a Notta's nodule [1]. Incidence of trigger finger is 2-3% and increase up-to 10% in diabetic patients [2]. Still exact causes have not been elucidated but local trauma, stress and excessive movement are the proposed theories. Various modalities for treatment of trigger thumb, includes Nonsteroidal anti-inflammatory drugs (NSAIDs), thumb splinting, passive extension exercises, steroid injections. Surgical intervention includes open and percutaneous A1 pulley release but some complications occur as infection, digital radial nerve injury, scar tenderness and joint contractures [3]. As per the sign, symptoms and pathogenesis of trigger thumb, it can be correlated to *Snayugata Vata* (vitiated *Vata* in tendons/ligaments). *Snayugata Vata* presents with *Stambha* (stiffness), *Shula* (pain), and *Akshepana* (inability of movement) [4]. *Snayu-Pradoshaja Vikara* (diseases of tendons/ligaments) as *Stambha* (stiffness), *Sankocha* (contraction), *Granthi* (nodule

like structure), and *Sphurana* (twitching) [5]. *Snayugata Vata* may be associated with acute or subacute conditions affecting ligaments and tendons, whereas *Snayu-Pradoshaja Vikara* may reflect their chronic manifestations. Here, *Snayugata Vata* and *Snayu-Pradoshaja Vikara* can be referenced to emphasize its clinical resemblance to trigger thumb. *Snayu-Pradoshaja Vikara* and *Snayugata vata* treated as per the principles of *Vatavikara*, particularly with *Snehana* (oleation), *Upanaha* (poultice), *Agnikarma*, (therapeutic heat), *Bandhana* (bandaging) and *Mardana* (massaging) [6]. Various Ayurvedic modalities were used to treat TT (trigger thumb) such as *Agnikarma* with honey, *Jalaukavcharan*, *Panchkarma* and internal medicines. *Agnikarma* helps people with a variety of painful musculoskeletal conditions, including *Snayugata vata* (w.s.r.to tennis elbow) [7], *Sandhigata vata* (w.s.r. to osteoarthritis in the knee) [8] and *Vatakantaka* (plantar fasciitis) and many more diseases. There are no previous studies available on the use of *Agnikarma* with *Tamra Shalaka* (Copper rod) for trigger thumb. In this case, the procedure is minimally invasive, cost effective, less time consuming and provide rapid relief from pain and locking, improving the condition from Grade 3 to Grade 1 in just three sittings (14 days) (see Table 4).

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Table 1
Local examination findings.

Examination	Findings
VAS scale	8/10
Tenderness	Present with palpable nodule at base of thumb (at 1st MCP joint)
Locking	Present with difficulty in extending the thumb, active flexion, passive extension
Grade (acc. to Quinell Grading System for Flexion and Extension)	Grade 3 severity trigger thumb right hand

2. Patient information

2.1. Case report

A male patient aged 56 years old came to the NIA hospital in the outdoor patient department (OPD) of *Shalya Tantra* (Surgery) in September 2024. He had complaints of aggravated pain and locking in right thumb since last 15 days. Patient was being working in printing press and having gradually progressive pain and intermittent locking of right thumb since last 6 months. There was no other past history of trauma to the affected right thumb. No any relevant medical or surgical history. Patient had taken conservative treatment such as anti-inflammatory and analgesic medicines at private Orthopedic hospital for the same in last 6 months. Patient didn't get long term relief in the symptoms of trigger thumb. After that he came to our hospital.

Table 2
Timeline of the case report.

Date	Event	Details				
March 2024	Starting of the complaints	As per patient, he developed mild pain and stiffness in right thumb, followed by development of intermittent locking of the thumb, then he took treatment from a private hospital but got mild relief in pain.				
August 2024	Aggravation of symptoms	As per patient, the pain and stiffness increased with passage of time and patient was not able to extend his right thumb without support.				
With these complaints patient visited our hospital.						
Day1 (10-09-2024)		Patient presents with pain, stiffness and locking of right thumb while working. Clinical examination confirmed trigger thumb.				
Initial consultation and first sitting of Agnikarma		1st sitting of Agnikarma therapy done after clinical assessment.				
		Oral medicines prescribed for 7 days -Dashamoola Kwatha (decoction) 20 ml twice a day after meal and Tablet Trayodashang Guggulu 500 mg twice a day after meal with lukewarm water, Tablet Sanjivani Vati 250 mg twice a day after meal with lukewarm water				
		Clinical assessment				
		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		8/10	Present over the flexor tendon sheath at the MCP joint	Present and requires passively correction	Present	3
Day 8(17-09-2024)		Second sitting of Agnikarma done after clinical assessment. Same oral medicines continued for next 7 days.				
Second Agnikarma Sitting		Clinical assessment				
		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		2/10	Reduced	Present but actively correctable	Present	2
Day 15 (25-09-2024)		Third and last sitting of Agnikarma done after clinical assessment. Same oral medicines continued for next 15 days.				
Third Agnikarma Sitting		Clinical assessment				
		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		1/10	No tenderness	No locking	Present size reduced	1
Day 45(25-10-2024)		Clinical assessment				
1st follow up		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		No Pain	No tenderness	No locking	Present size reduced	0
Day 75(24-11-2024)		Clinical assessment				
2nd follow up		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		No Pain	No tenderness	No locking	Present size reduced	0
Day 180(24-03-2025)		Clinical assessment				
3rd follow up		Vas (Visual Analogue Scale)	Tenderness	Locking	Nodule	Grade
		No Pain	No tenderness	No locking	Present size reduced	0

2.2. Clinical findings

Inspection revealed no swelling at the base of the right thumb. Active flexion of the right thumb by the patient resulted in thumb locking at the metacarpo-phalangeal (MCP) joint. The patient used another hand to passively extend the right thumb. On palpation, tenderness and a small palpable nodule was noted over the base of the MCP joint in the right thumb. Sensory and motor examinations of both hands were normal. On the basis of clinical examination, the patient was diagnosed with a trigger thumb (grade 3 severity according to the Quinell Grading System for Flexion and Extension) [9]. On general examination, the patient was fit and well oriented. All the vitals and routine blood investigations were within the normal range (Table 1).

3. Timeline of the case

4. Diagnostic assessment

On the basis of clinical examination, the patient was diagnosed with a trigger thumb right hand (grade 3 severity - Quinell Grading System for Flexion and Extension) [9].

5. Therapeutic intervention

The procedure of *Agnikarma* described to the patient and written consent taken before each sitting, three sittings of *Bindu* type (round

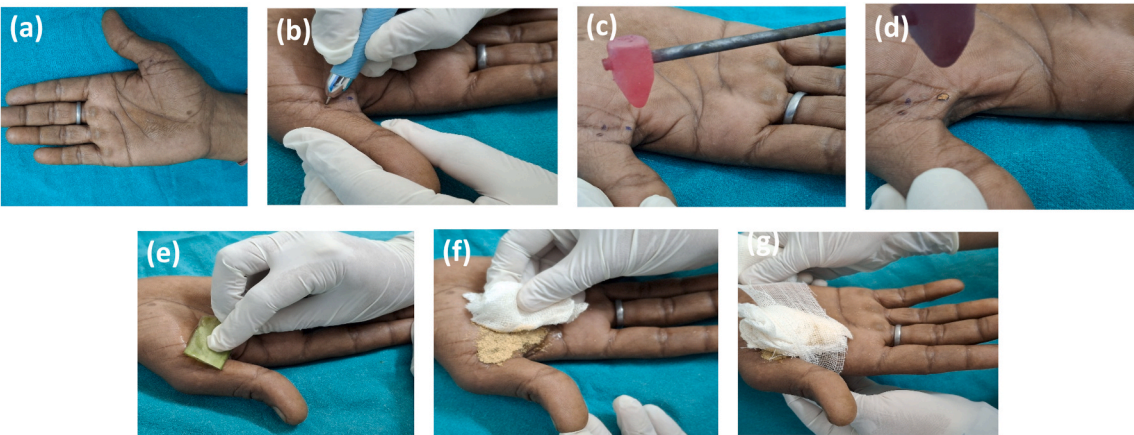


Fig. 1. Steps of Agnikarma procedure (a) Antiseptic cleaning, (b) Marking of most tender points, (c) Red hot *Tamra Dhatu Shalaka*, (d) Agnikarma at marked areas, (e) Application of *Aloe vera* over *Dagdha Sthana*, (f) Application of *Madhuyasthi Churna* (g) Bandaging.

Table 3
Duration of Agnikarma procedure.

Procedure step	Time taken
Pre-procedure preparation	Part preparation and marking tender spots-5 minutes
Main procedure	Heating <i>Tamra Shalaka</i> up to red hot- 10 minutes Application of <i>Shalaka</i> for 2 sec at each marked spot (3), total for 6 seconds.
Post procedure steps	Application of <i>aloe vera</i> , <i>Madhuyasthi Churna</i> and bandaging- 5 minutes
Total duration	15 minutes

shape) Agnikarma upto *Twaka Dagdha* with *Tamra Shalaka* (copper rod) done at the base of the right thumb on most tender spots at regular intervals of 7 days in outdoor patient department of the hospital.

5.1. Agnikarma 1st sitting

Agnikarma was performed in three stages, i.e.

- i) *Purvakarma* (pre procedure measures),
- ii) *Pradhankarma* (main procedure),
- iii) *Paschatkarma* (post procedure measures)

5.1.1. *Purvakarma*

The patient advised to take breakfast before Agnikarma and informed written consent was taken. Sterilized gloves, *Tamra Shalaka*, *Aloe vera* pulp, *Madhuyasti* (*Glycyrrhiza glabra* L.) *Churna* (powder), kidney tray, gas burner, sterile gauze, and bandages all kept ready for use during Agnikarma.

Table 4
Clinical assessment of the Trigger thumb.

	Agnikarma			After 1 month of 3rd sitting	After 2 months of 3rd sitting	After 6 months of 3rd sitting
	1st sitting	2nd sitting	3rd sitting			
Vas (Visual Analogue Scale)	8/10	2/10	1/10	No Pain	No Pain	No Pain
Tenderness	Present over the flexor tendon sheath at the MCP joint	Reduced	No tenderness	No tenderness	No tenderness	No tenderness
Locking	Present and requires passively correction	Present but actively correctable	No locking	No locking	No locking	No locking
Nodule	Present	Present	Present size reduced	Present size reduced	Present size reduced	Present size reduced
Grade	3	2	1	0	0	0

5.1.2. *Pradhanakarma*

The patient was placed in supine position. *Tamra Shalaka* heated on gas burner. Local site examined and area of maximum tenderness palpated and marked [Fig. 1(a), (b)]. Three *Bindu* type of Agnikarma (burn) were performed with red hot tip of *Tamra Shalaka* on marked points at around 5 millimeter (mm) distance [Fig. 1(c) and (d)].

5.1.3. *Paschatkarma*

Aloe vera pulp applied on *Dagdha Sthana* (burned area) for few seconds to minimize the burning sensation [Fig. 1(e)] after that *Madhuyasti Churna* (Liquorice powder) applied [Fig. 1(f)] and covered with sterile gauze followed by bandaging [Fig. 1(g)]. Bandaging kept for the whole day.

5.1.4. Duration of Agnikarma procedure

5.2. Agnikarma second sitting

The patient was re-examined for tenderness, stiffness, nodule, and locking of the thumb. The patient reported that pain and stiffness reduced, but thumb locking and nodule was still there. The previous *Dagdha Vrana* (burned wound) was found healed. After taking written consent, Agnikarma performed with same procedure as in first sitting.

5.3. Agnikarma third sitting

On examining, there was no tenderness, stiffness, and locking of the thumb, but still Notta's nodule was there. The previous *Dagdha Vrana* was found healed. The area of the nodule was palpated and marked. Three *Bindu* type of Agnikarma were performed at distance of 5 mm with *Tamra Shalaka* at marked points. The rest of all the steps were followed

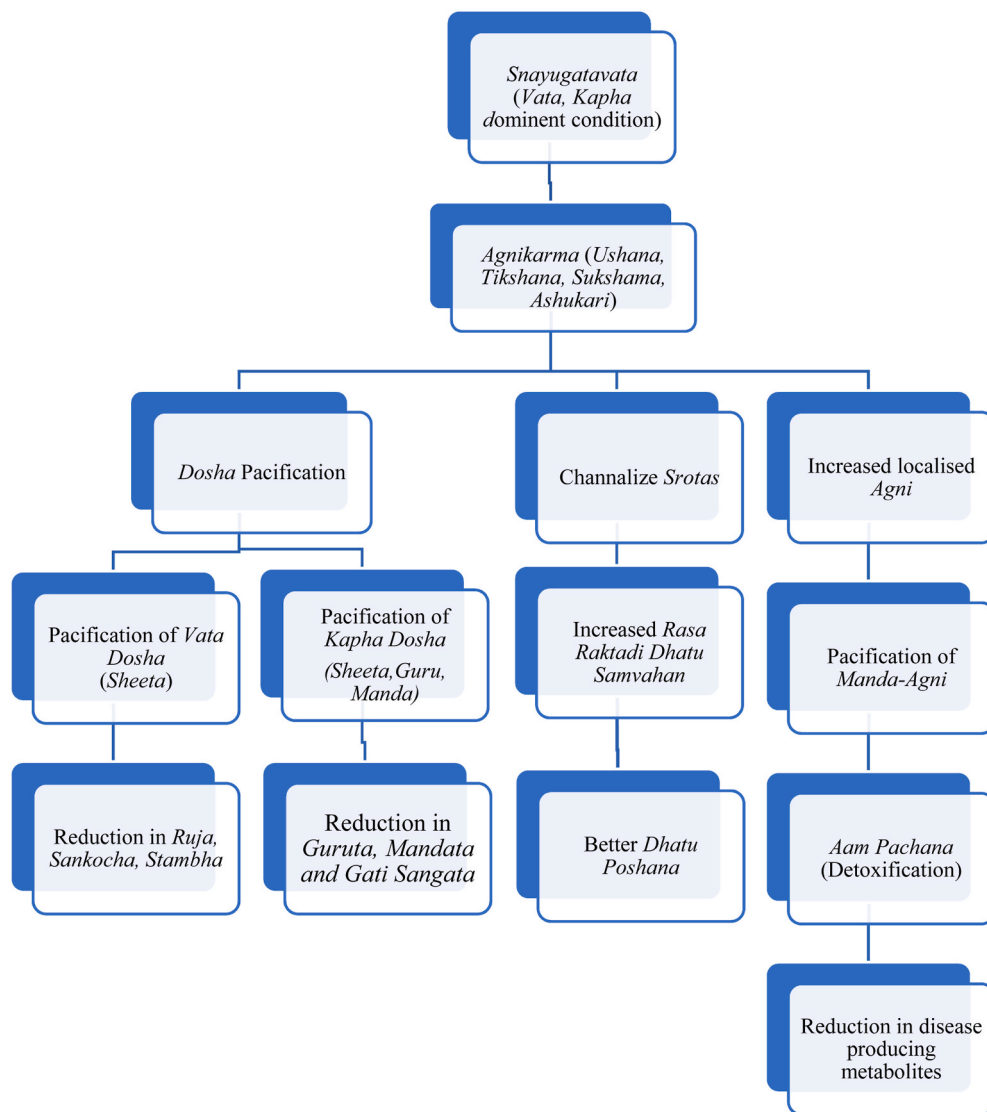


Fig. 2. Probable mode of action of Agnikarma according to Ayurveda.

as mentioned in the first sitting.

5.4. Oral medication

The patient was also prescribed *Dashamoola Kwatha* (decoction) 20 ml twice a day after meal and Tablet *Trayodashang Guggulu* 500 mg twice a day after meal with lukewarm water, Tablet *Sanjivani Vati* 250 mg twice a day after meal with lukewarm water. These oral medicines were given for 1 month.

6. Follow up and outcomes

The patient was assessed on every sitting for change in signs and symptoms. There was no recurrence of disease during the 6 months of follow up period (Table 4).

7. Discussion

TT is a musculoskeletal disease of the thumb and also called stenosing tenosynovitis. As it is considered as *Snayugata Vata*, having vitiated *Vata* and *Kapha* with symptoms like pain, stiffness, and thumb locking. *Agnikarma* is effective to pacify the *Vata-Kapha Dosha* by its *Ushna* (hot), *Sukshma* (penetrating), and *Ashukari* (fast-acting)

qualities. It channelizes the *Srotas* (channels), stimulate tissue metabolism (Fig. 2) [10]. In this case *Agnikarma* helped to relieve pain and stiffness after the first and second sittings (Table 3) and the locking was also subsided after third sitting of *Agnikarma*. As per modern literature, the production of type III collagen increases at the inner layer of the A1 pulley due to friction with the flexor tendon. These collagens melt at more than 50 °C temperature [11]. Heat shock proteins (HSPs) are produced in response to stress from local heat application, which stimulates the production of heat shock proteins (HSPs). These proteins, in turn, trigger the release of anti-inflammatory cytokines, helping to reduce inflammation and pain (Fig. 3) [12]. Possibly, *Agnikarma* helps to slow down the collagen production at the A1 pulley, leading to the release of stiffness, pain, and locking of the thumb. It requires further research to explore the action on the A1 pulley, flexor tendon, and its adjoining structures [13]. *Trayodashang Guggulu* is having properties to treat *Vata* and *Kapha* dominant conditions as *Sandhi* (joint), *Ashti* (bone), *Majja* (bone marrow), *Snayu* (ligaments/tendons) *Ashrita Vata* [14]. *Dashmool kwatha* have properties of pacifying *Tridosha*, [15] *Shothhara* (anti-inflammatory action) [16]. *Sanjivani Vati* is a diaphoretic and detoxifying formulation that reduces *Aama* (toxins) and may clear *Srotas* (channels) blockages, and by these actions it reduces stiffness [17]. These oral medications have systemic anti-inflammatory action and *Agnikarma* have local effect. It requires further research to explore the

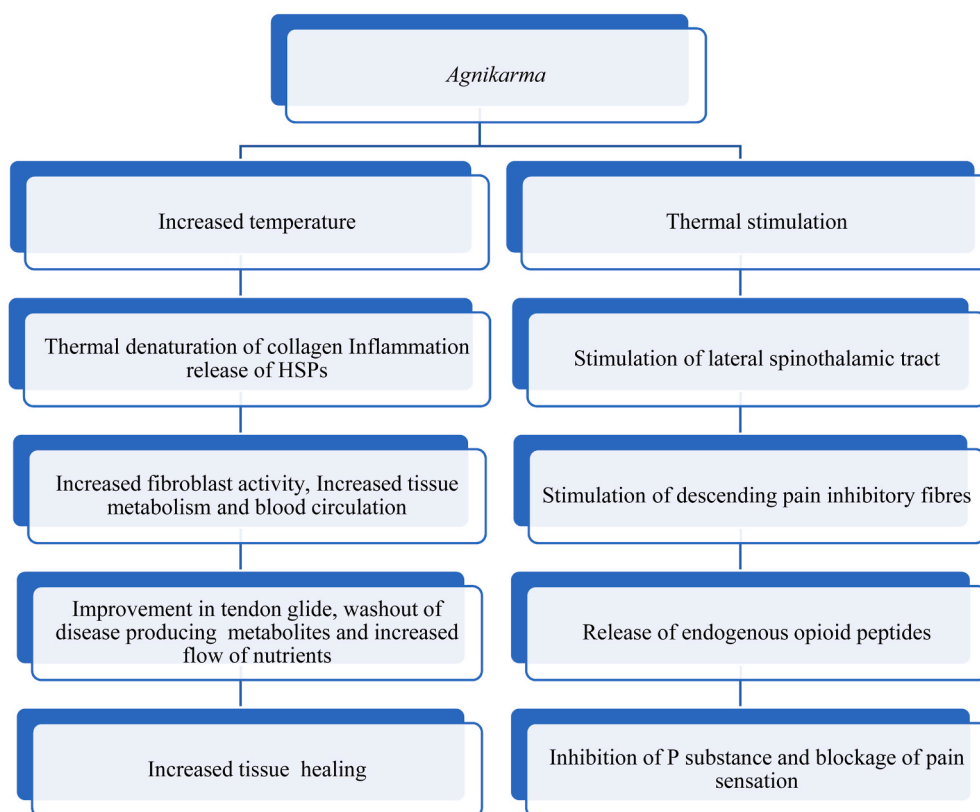


Fig. 3. Probable mode of action of *Agnikarma* as per modern principles.

action on *Agnikarma* on A1 pulley, flexor tendon and its adjoining structures. Oral medications are also working as an adjuvant with *Agnikarma* in Trigger thumb.

8. Further research needed

A single case report can't authenticate the efficacy of *Agnikarma*, so it requires randomized control trials with different kinds of *Shalaka*. The exact action of *Agnikarma* on trigger thumb is to be explored.

9. Conclusion

This case highlights significant clinical improvement, particularly in the symptoms of trigger thumb, which severely affected the patient's daily functioning and mental well-being. This condition was managed by using *Agnikarma* therapy, which is safe, cost-effective, and minimally invasive procedure, in combination with oral Ayurvedic medications. Notably, the patient has reported no recurrence or new complaints to date. This outcome underscores the strength and uniqueness of this case report.

10. Patient perspective

"I'm a 56-year-old man, I was having pain in right hand thumb which occasionally locked in a bent position. I was not able to straighten the thumb without support. It was affecting my daily routine activities. I came to hospital and underwent *Agnikarma* therapy and some medications. Now I don't have any pain in thumb and I can easily move the thumb without any locking. Now I can perform my routine without any difficulty in thumb" (24-03-2025).

Informed consent

Informed consent from patient taken for publication.

Author contributions

BS: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, original draft manuscript preparation, Follow Up.

SS: Methodology, Validation, Resources, Data curation, Reviewing and Editing, Visualisation, Supervision.

MS: Methodology, Validation, Resources, Data curation, Reviewing and Editing, Visualisation, Supervision.

RS: Methodology, Validation, Resources, Data curation, Reviewing and Editing, Visualisation, Supervision.

PG: Conceptualization, Methodology, Formal analysis, Investigation, Original draft manuscript preparation, Data curation.

Declaration of generative AI in scientific writing

We declare that we have not used any writing tool/service or AI for scientific writing.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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